

Project 4: Beam Training with Analog True-Time-Delay Arrays

Beam training in a wideband millimeter-wave (mmWave) system with conventional arrays based on phase shifters requires exhaustive beam sweeping to find the dominant propagation directions in the channel. With such type of arrays, all frequency components of the signal are steered to or combined from the same direction because the same phase shifts are applied to all signal frequencies. Recent advances in the design of true-time-delay (TTD) arrays allowed their adoption in wideband mmWave systems. Compared to the arrays based on phase shifters, TTD arrays have the ability to synthesize frequency-dependent beams by delaying the signal in all antenna array branches. A user with a TTD array can combine different frequencies from different angular directions at the same time and thus conclude the dominant propagation direction in a fast way using simple frequency-domain digital signal processing (DSP). In this project, you will study TTD-based beam training on the user side. Specifically, you will design a frequency-dependent beam training codebook for analog TTD arrays, implement a dictionary-based angle estimation algorithm, and study the impact of hardware impairments on beam training performance.

- **System model:** Consider the orthogonal frequency division multiplexing (OFDM) based system model described in [R-5, Sec. II]. The base station is assumed to be equipped with a digital array. It uses a fixed precoder $\mathbf{v}$, which is designed according to a known angle of departure and it is the same for all subcarriers. The user is assumed to have a TTD array with delay elements implemented in baseband domain and frequency-dependent beams described in [R-5, Sec. II-A].
- **Channel model:** Implement a wideband channel model with $L = 3$ multipath components, where the dominant path is 10 dB stronger than each of the other two. The channel model is described in [R-5, Sec. II].
- **TTD codebook design:** Design frequency-dependent codebooks for analog TTD arrays, as explained in [R-5, Sec. III]. To set up a codebook, you need to design delay and phase taps in all antenna branches.
- **Angle estimation algorithm:** Implement the dictionary-based estimation algorithm described in [R-5, Sec. IV] and summarized in [R-5, Sec. IV, Alg. 1].
- **Impact of hardware impairments:** Study the impact of hardware impairments, including the gain and phase errors, on the beam training performance with analog TTD arrays. The models of delay and phase errors are described in [R-5, Sec. II-A], while their impacts on TTD combiners are included in [R-5, (5), second equation].
- **Results:** Study beam training with analog TTD arrays. Specifically, visualize the designed training codebook and evaluate the root mean square error (RMSE) of angle of arrival estimation. Assume the system parameters as in [R-5, Sec. V].
 1. Plot the beam patterns of designed codebook for analog TTD arrays.
 2. Plot the RMSE of angle estimation vs. Signal-to-Noise Ratio (SNR) in the range of -20 dB to 20 dB for diversity factor $R = 1, 2, 4$, in the absence of hardware impairments (i.e. without delay, gain and phase errors).

3. Plot the RMSE of angle estimation vs. standard deviation of gain error σ_A in the range 0 dB to 4.5 dB. Refer to the red dashed line with stars in [R-5, Fig. 4] for comparison with your results.
4. **Bonus:** Plot the RMSE of angle estimation vs. standard deviation of phase error σ_P in the range 0° to 50° . Refer to the red dashed line with diamonds in [R-5, Fig. 4] for comparison with your results.

Relevant References

[R-5] V. Boljanovic, H. Yan, E. Ghaderi, D. Heo, S. Gupta and D. Cabric, “Design of Millimeter-Wave Single-Shot Beam Training for True-Time-Delay Array,” 2020 IEEE 21st International Workshop on Signal Processing Advances in Wireless Communications (SPAWC), pp. 1-5, 2020 (IEEE Xplore link)